



Kunal J
Data Scientist
Kunaljanjirala6@gmail.com
225-493-8348

PROFESSIONAL SUMMARY

- Professional qualified **Data Scientist/Data Analyst** with **around 5 years** of experience in **Data Science and Analytics including Data Mining, Deep Learning/Machine Learning, Azure and Statistical Analysis**
- Expert in data integration and statistical analysis, transforming complex business challenges into actionable insights using **Python, SQL, and cloud platforms**.
- Expertise in data integration and modeling across disparate systems, with proven ability to uncover complex insights using advanced database technologies.
- Good understanding of **Neo4j graphical databases** to design the graphical structure applications.
- Skilled in translating intricate analytical challenges into structured problem-solving approaches, leveraging **graph database** techniques for enhanced **data exploration and pattern recognition**.
- Effectively developed models from end-to-end: preprocessing, feature engineering, training, validation, **hyperparameter tuning**, and model deployment.
- Translated business challenges into data-driven solutions through the utilization of advanced analytics techniques and methodologies.
- Skilled in developing data-driven solutions for logistics, leveraging advanced statistical modeling and machine learning techniques to reduce operational costs and improve service quality.
- Expertise in leveraging statistical modeling and machine learning algorithms to uncover **data patterns, trends, and actionable insights**.
- Experienced with machine learning algorithm such as logistic regression, **KNN, SVM**, random forest, neural network, NLP, linear regression, lasso regression and k-means.
- Experience in building data pipelines using **Azure Data factory, Azure data bricks** and loading data to **Azure data Lake, Azure SQL Database, Azure SQL Data warehouse** and controlling and granting database access.
- Skilled in managing **Active Directory, Azure/EntraID, and ARS**, with a focus on user accounts, groups, **SSO, SAML, and MFA**.
- Proficient in **predictive analytics** and **machine learning** for logistics, including demand forecasting, route optimization, and fraud detection.
- Implemented **Bagging** and **Boosting** to enhance the model performance.
- Co-authored a research paper on understanding the code-generating performance of ChatGPT, contributing to the advancement of **Large Language Models (LLMs)** in the AI community.
- Experience in implementing data analysis with various analytic tools, such as Anaconda Jupiter Notebook 4.X, R 3.0 (ggplot2, dplyr, Caret) and Excel.
- Solid ability to write and optimize diverse SQL queries, working knowledge of **RDBMS** like SQL Server, NoSql databases like MongoDB.

- Design, develop, and deploy machine learning models using **GCP's BigQuery, Google Cloud AI Platform,** and AutoML.
- Excellent understanding **Agile** and **Scrum** development methodology
- Used the version control tools like Git 2.X and build tools like Apache Maven/Ant
- Experienced the full software life cycle in **SDLC, Agile, DevOps** and **Scrum methodologies** including **creating requirements, test plans.**
- Skilled in Advanced **Regression Modeling, Correlation, Multivariate Analysis, Model Building, Business Intelligence tools** and **application** of Statistical Concepts.
- Developed predictive models using **Decision Tree, Naive Bayes, Logistic Regression, Random Forest, Social Network Analysis, Cluster Analysis,** and **Neural Networks.**
- Experienced in Machine Learning and Statistical Analysis with **Python Scikit-Learn.**
- Strong SQL programming skills, with experience in working with **functions, packages** and **triggers.**
- Expertise in transforming business requirements into **designing algorithms, analytical models, building models, developing data mining** and **reporting solutions** that scales across massive volume of structured and unstructured data.
- Skilled in performing **data parsing, data manipulation, data architecture, data ingestion and data preparation** with methods including describe data contents, compute descriptive statistics of **data, regex, split and combine, merge, Remap, subset, reindex, melt** and **reshape.**
- Worked with **NoSQL Database** including **Hbase, Cassandra** and **MongoDB.**
- Experienced in **Big Data** with **Hadoop, MapReduce, HDFS** and **Spark.**
- Experienced in Data **Integration Validation** and **Data Quality controls** for **ETL process** and **Data Warehousing** using **MS Visual Studio, SSAS, SSIS** and **SSRS.**
- Automated recurring reports using **SQL and Python** and visualized them on **BI platform** like **Tableau.**
- Excellent communication skills. Successfully working in fast-paced multitasking environment both independently and in collaborative team, a self-motivated enthusiastic learner.

TECHNOLOGIES:

BigData/Hadoop Technologies	Hadoop, HDFS, YARN, MapReduce, Hive, Pig, Impala, Sqoop, Flume, Spark, Kafka, Storm, Drill, Zookeeper and Oozie Giraph (for graph processing in Hadoop), Apache TinkerPop
Languages	HTML5,DHTML, WSDL, CSS3 ,C, C++, XML,R/R Studio, SAS Enterprise Guide, SAS ,R (Caret, Weka, ggplot) , Perl, MATLAB, Mathematica, FORTRAN, DTD, Schemas, Json, Ajax, Java, Scala, Python (NumPy, SciPy, Pandas, Gensim, Keras), Java Script, Shell Scripting, Gremlin, Cypher.
NO SQL Databases	Cassandra, HBase, MongoDB, MariaDB, TigerGraph
Business Intelligence Tools	Tableau server, Tableau Reader, Tableau, Splunk, SAP Business Objects, OBIEE, SAP Business Intelligence, QlikView, Amazon Redshift, or Azure Data Warehouse, D3.js, Gephi
Development Tools	Microsoft SQL Studio, IntelliJ, Eclipse, NetBeans.
Development Methodologies	Agile/Scrum, UML, Design Patterns, Waterfall
Build Tools	Jenkins, Toad, SQL Loader, Maven, ANT, RTC, RSA, Control-M, Oziee, Hue, SOAP UI

Reporting Tools	MS Office (Word/Excel/Power Point/ Visio/Outlook), Crystal reports XI, SSRS, cognos 7.0/6.0, Graphana.
Databases	Microsoft SQL Server 2008,2010/2012, MySQL 4.x/5.x, Oracle 11g, 12c, DB2, Teradata, Netezza
Operating Systems	All versions of Windows, UNIX, LINUX, Macintosh HD, Sun Solaris
Graph Analytics	GraphX (Apache Spark graph processing), GraphFrames (Apache Spark graph processing), Giraph (for graph processing on Hadoop), Apache Flink (streaming graph analytics).

PROFESSIONAL EXPERIENCE

Client: Home Depot, Atlanta, GA
Data Scientist

Oct 2023 - Present

Responsibilities:

- Extracted data from **HDFS** and prepared data for exploratory analysis using **data munging**.
- Implemented machine learning models to predict delivery delays and **optimize logistics routes, using Random Forests and Gradient Boosting algorithms**.
- Designed a decision tree application in **Neo4j** to model and visualize logistics operations, enabling better decision-making for **transportation management**.
- Conducted **feature engineering on logistics datasets** to extract key patterns such as delivery time trends, demand peaks, and vehicle utilization rates.
- Built models using Statistical techniques like Bayesian HMM and Machine Learning classification models like **XG Boost, SVM, and Random Forest**.
- Built machine learning models using **Neo4j** and **GraphX** to analyze complex relationships and improve pattern recognition for tasks like fraud detection and recommendations.
- Identified relationship between different data sets in **GraphDB** database.
- Utilized advanced machine learning techniques such as **Hyperparameter Tuning** and **Model Experimentation** to optimize model performance.
- Engineered features and processed complex datasets using Databricks to identify **missing data, anomalies, and outliers** for robust predictive modeling.
- Conducted feature engineering to enhance model accuracy, focusing on **extraction, transformation, and selection of critical variables**.
- Designed and implemented **Data Lineage graph** from origination to consumption point of view
- Used **Gephi** and **Tableau** (with Neo4j connector) to create interactive graph visualizations, enhancing data insights for decision-making.
- Utilized Python and Spark to preprocess large logistics datasets, ensuring data accuracy and consistency for effective model development.
- Conducted **Geospatial Data Analysis** using Python and Neo4j to visualize logistics hotspots, improving delivery coverage and decision-making.
- Participated in all phases of **data mining, data cleaning, data collection, developing models, validation, visualization** and **performed Gap analysis**.
- Designed and developed a decision tree application using Neo4J graph database to model the nodes

and relationships for each decision.

- Developed and deployed machine learning solutions for logistics **anomaly detection**, including identifying package handling errors and **fraud risks**.
- Presented logistics data insights through **interactive dashboards and visualizations**, aiding stakeholders in improving **supply chain strategies**.
- A highly immersive Data Science program involving **Data Manipulation & Visualization, Web Scraping, Machine Learning, Python programming, SQL, GIT, MongoDB, Hadoop**.
- Setup storage and data analysis tools in AWS **cloud computing infrastructure**.
- Installed and used Caffe **Deep Learning Framework**
- Worked on different data formats such as **JSON, XML** and performed machine learning algorithms in Python.
- Fine-tuned large language models (**LLMs**) like GPT and T5 for domain-specific applications, improving text generation accuracy and contextual relevance.
- Developed and deployed multimodal Gen AI solutions combining text, image, and video generation using frameworks like **Hugging Face and Stable Diffusion**.
- Used Python files to load the data from CSV files to **Neo4J** graphical database
- Worked with APIs like OpenAI to embed guardrails in **LLM workflows**.
- Experience working with Data Frames, RDD, Spark SQL, Spark Streaming, APIs, System Architecture, and Infrastructure Planning.
- Implemented Retrieval-Augmented Generation (**RAG**) models to enhance the knowledge base of the AI systems, enabling them to generate more informed and contextually relevant outputs.
- Design, develop, and deploy machine learning models using **Google Cloud AI Platform and AutoML**.
- Worked as Data Architects and IT Architects to understand the movement of data and its storage and ER Studio 9.7
- Used **pandas, numpy, seaborn, matplotlib, scikit-learn, scipy, NLTK** in **Python** for developing various machine learning algorithms.
- Data Manipulation and Aggregation from different source using **Nexus, Business Objects, Toad, Power BI and Smart View**.
- Implemented **Agile Methodology** for building an internal application.
- Focus on integration overlap and Informatica newer commitment to **MDM** with the acquisition of Identity Systems.
- Coded proprietary packages to analyze and visualize **SPC** file data to identify bad spectra and samples to reduce unnecessary procedures and costs.
- Programmed a utility in Python that used multiple packages (**numpy, scipy, pandas**)
- Implemented Classification using supervised algorithms like **Logistic Regression, Decision trees, Naive Bayes, KNN**.
- As Architect delivered various complex **OLAP databases/cubes, scorecards, dashboards and reports**.
- Updated **Python** scripts to match training data with our database stored in **AWS Cloud Search**, so that we would be able to assign each document a response label for further classification.
- Used Teradata utilities such as **Fast Export, MLOAD** for handling various tasks data **migration/ETL** from **OLTP Source Systems to OLAP Target Systems**
- Developed Python scripts to automate and provide Control flow to **Pig scripts**.
- Data transformation from various resources, data organization, features extraction from raw and stored.
- Validated the machine learning classifiers using **ROC Curves and Lift Charts**.

Environment: Unix, Python, MLLib, SAS, regression, logistic regression, Hadoop, NoSQL, AI, Teradata, OLTP, random forest, OLAP, HDFS, ODS, NLTK, SVM, JSON, XML and MapReduce.

Client: TD Bank; New Jersey
Data Scientist

Oct 2022 - Sept 2023

Responsibilities:

- Utilized Spark, **Scala, Hadoop, HQL, VQL, oozie, pySpark, Data Lake, TensorFlow, HBase, Cassandra, Redshift, MongoDB, Kafka, Kinesis, Spark Streaming, Edward, CUDA, MLLib, AWS, Python**, a broad variety of machine learning methods including classifications, regressions, dimensionally reduction etc.
- Utilized the engine to increase user lifetime by 45% and triple user conversations for target categories.
- Application of various machine learning algorithms and statistical modeling like **decision trees, text analytics**, natural language processing (**NLP**), supervised and unsupervised, regression models, social network analysis, neural networks, deep learning, **SVM**, clustering to identify Volume using **scikit-learn** package in **python, Matlab**.
- Perform **NLP analysis** to determine which N number of topics are important for the disposition.
- Categorize each note into one of the N number of topics above using a document-matching NLP methodology.
- Used Natural Language Processing (NLP) for response modeling and fraud detection efforts for credit cards.
- Involved in converting Hive/SQL queries into Spark transformations using Spark RDD, Scala and Python.
- Worked on analyzing **data from Google Analytics, AdWords, Facebook etc.**
- Evaluated models using Cross Validation, Log loss function, **ROC** curves and used **AUC** for feature selection and elastic technologies like **ElasticSearch, Kibana**.
- Performed Data Profiling to learn about behavior with various features such as traffic pattern, location, Date and Time etc.
- Developed Spark code using Scala and **Spark-SQL/Streaming** for faster testing and processing of data
- Generation (**RAG**) for various generative **AI** models. Designing and implementing information retrieval systems using embeddings, vector stores.
- Categorized comments into positive and negative clusters from different social networking sites using Sentiment Analysis and Text Analytics.
- Experience in using Maven for building and deploying **J2EE Application** archives (Jar and War) on Web Logic, **IBM Web Sphere**.
- Hands on experience with build and deploying tools like **Maven** and GitHub using **Bash scripting**.
- Involved in configuring builds using **Jenkins with Git** and used Jenkins to deploy the applications onto Dev, QA environments.
- Performed **Multinomial Logistic Regression, Decision Tree, Random Forest, SVM** to classify package is going to deliver on time for the new route.
- Performed data analysis by using Hive to retrieve the data from **Hadoop cluster, Sql** to retrieve data from Oracle database and used **ETL** for data transformation.
- Performed Data Cleaning, features scaling, features engineering using **pandas** and **numpy** packages in **python**.
- Exploring DAG's, their dependencies and logs using AirFlow pipelines for automation
- Performed data cleaning and feature selection using **MLlib** package in **PySpark** and working with deep learning frameworks such as **Caffe, Neon**.

- Developed **Spark/Scala, R Python** for regular expression (regex) project in the **Hadoop/Hive** environment with **Linux/Windows** for big data resources.
- Managed **Active Directory, Azure/EntraID, and ARS**, automating user management and **MFA** policies using PowerShell and Python.
- Implemented **SSO, SAML, and OAuth** for secure access control and integration across applications, ensuring confidentiality.
- Used clustering technique **K-Means** to identify outliers and to classify unlabeled data.
- Tracking operations using sensors until certain criteria is met using **AirFlow technology**.
- Analyze traffic patterns by calculating autocorrelation with different time lags.
- Ensured that the model has low False Positive Rate and Text classification and sentiment analysis for unstructured and semi-structured data.
- Addressed over fitting by implementing algorithm regularization methods like **L1 and L2**.
- Used Principal Component Analysis in feature engineering to analyze high dimensional data.
- Used **MLlib, Spark's Machine learning** library to build and evaluate different models.
- Implemented rule-based expertise system from the results of exploratory analysis and information gathered from the people from different departments.
- Created and designed reports that will use gathered metrics to infer and draw logical conclusions of past and future behavior.
- Developed **MapReduce** pipeline for feature extraction using **Hive** and **Pig**.
- Created Data Quality Scripts using **SQL** and **Hive** to validate successful data load and quality of the data. Created various types of data visualizations using **Python** and **Tableau**.
- Communicated the results with operations team for taking best decisions.
- Collected data needs and requirements by Interacting with the other departments.

Environment: Python 2.x, CDH5, HDFS, Hadoop, Hive, Impala, AWS, Linux, Spark, AI, Tableau Desktop, SQL Server, Microsoft Excel, Matlab, Spark SQL, Pyspark.

Client: Accenture; Hyderabad, India

July 2020 – June 2022

Data Scientist/ Data Analyst

Responsibilities:

- Created new features based on information from million transaction records and training models using **Machine-Learning** techniques such as Gradient **Boosting Tree** and **Deep Learning**.
- Analyzed and determined a cutoff point for accepting/ declining transactions to minimize fraud losses and increase customer experience by using various machine learning algorithms such as Logistic Regression, Classification, Random Forests and Clustering in **SAS, R and Python**.
- Used **Pandas, Numpy, Seaborn, Scipy, Matplotlib, Scikit-learn, NLTK** in **Python** for implementing various machine learning algorithms.
- Used **SAS, SQL, Oracle, Teradata** and **MS Office** analysis tools to complete analysis requirements. Created **SAS** data sets by extracting data from Oracle database and flat files
- Used **Proc SQL, Proc Import, SAS Data Step** to clean, validate and manipulate data.
- Performed updating data by weekly and monthly; maintained, manipulated the data for database management. Used the **SAS Macro** and **Excel Macro** for the monthly production.
- Experienced in **SQL** queries to retrieve and validate data, prepared for data mapping document.
- Experienced in creating dashboards using **SSAS** and building matrix and tabular reports using reporting services
- Worked on **RDBMS** like **MySQL** and **NoSQL** databases like **MongoDB**.

- Used the **Agile Scrum** methodology to build the different phases of software development life cycle.

Environment: SAS9.4, Base SAS, SAS Macros SAS Graph, SAS Access, SAS STAT, SAS ODS, SAS SQL, SAS/ETL, SAS/Stat, SAS ENTERPRISE Miner, Python, PL/SQL, Oracle 9i, Hadoop, MongoDB